

OIML R76 Report System — User Guide

What this system does

The OIML R76 Report System is a digital type-test report system for non-automatic weighing instruments. It lets you:

- Register instrument families and models
 - Create test projects and schedule test runs
 - Record measurements, observations, and reference weights
 - Auto-evaluate results against OIML R76 / EN 45501 MPE tolerances
 - Generate reports in HTML, DOCX, and print-ready PDF formats
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Getting started

Open the app

Open the Railway domain in any modern browser (Chrome, Firefox, Edge). You will see the login screen.

Sign in

Two demo accounts are seeded into the database:

Role	Email	Password
Admin	admin@r76.local	admin123
Technician	tech@r76.local	tech123

Use either account to sign in. After login you land on the Dashboard.

Dashboard

The Dashboard shows:

- **Summary cards** — total projects, open tests, passed/failed counts, recent reports
- **Recent projects** — click any project to open it
- **Quick filters** — date range presets (7d / 30d / 90d) or a custom range, plus ascending/descending sort

Use the date picker at the top-right to filter the data shown on the Dashboard.

Projects

Create a project

1. Click **New Project**.
2. Fill in the project form:
 - **Instrument family** — choose from the existing list (e.g. Class II, Class III, Class IIII).
 - **Instrument model** — choose a model registered under that family.
 - **Client / manufacturer** — optional free text.

- **Serial number / range** — optional identifiers.

3. Click **Save**. The project opens.

Project screen

Inside a project you see several tabs:

- **Overview** — project details, status, and quick actions.
- **Tests** — list of test runs, with date, type, and verdict.
- **Observations** — free-text notes attached to the project.
- **Reports** — generated reports for this project.

Add an instrument model

Instrument models belong to a family. If the model you need is missing:

1. Go to **Settings** → **Instrument Families** (or ask an admin).
 2. Select the family and click **Add model**.
 3. Enter the model name, class, and any relevant standards.
 4. Save. The model is now available in the project form.
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Tests

Create a test run

1. Open a project → **Tests** tab → **New Test**.
2. Choose the **test type** (e.g. repeatability, eccentricity, temperature influence).
3. Enter the **test date**.
4. Fill in the measurement table:
 - Each row is a measurement condition.
 - Enter values in the columns shown (load, indication, error, etc.).
 - Cells that are not applicable to the current test type are disabled automatically.
5. Click **Save**.

Test verdicts

The system evaluates each measurement against the MPE tolerance table selected for the project:

- **Pass** — error is within tolerance.
- **Fail** — error exceeds tolerance.
- **N/A** — condition not evaluated.

Verdicts are shown inline in the measurement table and summarised at the bottom of the test run.

Date filtering on the Tests tab

Use the date filter at the top of the Tests list:

- **Presets** — Last 7 days, Last 30 days, Last 90 days.
- **Custom** — pick a From and To date.
- **Sort** — newest first or oldest first.

The filter applies only to the test list; project overview is unaffected.

Observations

Add an observation

1. Open a project → **Observations** tab → **New observation**.
2. Enter a short title and the observation text.
3. Optionally attach a photo (tap the camera icon or upload a file).
4. Save. The observation appears in the list with its timestamp.

Photos are stored on the server. They are regenerated from the database if the server is redeployed.

Reports

Generate a report

1. Open a project → **Reports** tab → **Generate report**.
2. Choose the **format**:
 - **HTML** — opens in the browser, print-ready with A4 page rules.
 - **DOCX** — downloads a Word document with the same structure.
 - **PDF** — requests a rendered PDF. If a browser binary is available the server returns a real PDF; otherwise it returns the print-ready HTML file with instructions to use the browser's **Save as PDF**.
3. Click **Generate**.

The new report appears at the top of the reports list.

View and download

- Click the report row to open it in a new tab.
- Use the browser's print/save dialog to save a local copy.
- Reports are also listed on the global **Reports** page, where you can filter by date and format across all projects.

Report formats

Format	Behaviour
HTML	Always available. Paginated for A4.
DOCX	Always available. Opens in Word or compatible editors.
PDF	Real PDF when a Chrome binary is installed; otherwise print-ready HTML with Save-as-PDF guidance.

Global Reports page

The **Reports** page (top navigation) lists every generated report in the system:

- Filter by **project**, **format**, and **date range**.

- Sort newest or oldest first.
 - Click any row to open the report.
 - Use the search box to find reports by project name or number.
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Users and roles

Roles

Role	What they can do
Admin	Full access: manage users, instrument families, projects, tests, reports.
Technician	Create and run tests, add observations, generate reports. Cannot manage users or families.

Manage users (admin only)

1. Go to **Settings** → **Users**.
2. Click **New user**, fill in name, email, role, and initial password.
3. Save. The user can sign in immediately.

Change your password

1. Click your name (top-right) → **Profile**.
 2. Enter the current password and the new password twice.
 3. Save.
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Tips and known limits

- **Uploaded photos and generated reports live on the server's disk.** They disappear if the service is redeployed or restarted. Database rows remain, and files regenerate on demand.
 - **Real PDF rendering** requires a Chrome-compatible browser binary on the server. The Railway deployment ships without one; PDF requests return print-ready HTML instead.
 - **Session timeout** is 12 hours by default. After that you must sign in again.
 - **Date filters** apply independently on the Dashboard, Tests list, and Reports page — they do not sync across pages.
 - **Keyboard shortcut:** press Esc on any form to cancel and return to the previous screen.
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Troubleshooting

Sign-in fails with "Cannot reach the API"

This means the browser cannot talk to the server. Check:

1. The URL is correct (the Railway domain, no extra paths).
2. You are not behind a corporate proxy that blocks the domain.
3. The server is running (ask the administrator to check the Railway deploy log).

"Circuit breaker" or "too many authentication failures" on login

The server could not connect to the database. The administrator must check the DATABASE_URL in Railway Variables:

1. The URL must use the **Transaction** pooler (port 6543).
2. The password must match the database password set in the Supabase dashboard.
3. After fixing, wait two minutes for the circuit breaker to reset, then redeploy.

Reports show "Puppeteer is not installed"

PDF generation fell back to HTML. Open the file in your browser and use **File** → **Save as PDF** (or Ctrl+P / Cmd+P). The layout is already paginated for A4.

Photos missing after a redeploy

Redeploying the service wipes the uploads directory. The photo thumbnails in the UI will show as missing until the observation is re-edited and the photo re-attached. The database record of the observation is unaffected.